

SHUBHAM KALE

3510 N Charles St | 480-589-7636 | skale1@jh.edu

EDUCATION

Johns Hopkins University, Whiting School of Engineering

B.S. Applied Mathematics & Statistics, B.A. Economics | Expected 05/29 | Cumulative GPA: 3.70/4.00

Honors & Awards: Hodson Trust Scholar (selective merit scholarship, ~20/45,000 applicants), 1For2 Foundation Scholarship (~5/1,000); ~\$300K total scholarships awarded

MACHINE LEARNING & TECHNICAL SKILLS

Machine Learning: deep learning (PyTorch, CNNs), RNN/Transformer sequence models, learning-to-rank, retrieval-augmented generation (RAG), LLM evaluation & failure-mode analysis, ensemble methods (XGBoost), SHAP interpretability

Statistics & Methods: causal inference (G-formula), probabilistic & epidemiological modeling, experimental design, stratified sampling, hyperparameter search (GridSearchCV), model validation

Languages & Infra: Python (PyTorch, scikit-learn, pandas, NumPy), C/C++, R, SQL, FastAPI, AWS, Google Cloud, Git, CI pipelines

MACHINE LEARNING RESEARCH & EXPERIENCE

Summer Atlantic Capital LLC & SAC Nexus

Jun 2026 – Present

Product Development Team Member — Data Scientist

- Build the **ML core** of an **anomaly-detection** system as a versioned **FastAPI** microservice, owning the repository/CI scaffold and **event-ingestion pipeline**
- Develop a **temporal sequence model** (RNN/Transformer over event streams), a **learning-to-rank** prioritization model, and an **LLM explanation layer** with **retrieval-augmented generation**; document assumptions, limitations, and performance metrics for institutional use

Gates Foundation Health Equity Project | JHU PIER/PACER

Sep 2025 – Jun 2026

Lead Model Developer — under Dr. Kate Rucinski

- Trained **gradient-boosted classifiers** (XGBoost) with **SHAP-based feature selection** on ~10M-record national survey data to identify primary drivers of multimorbidity
- Designed **G-formula causal-inference** intervention simulations for the **Gates-funded** JHU–South African Medical Research Council partnership; coordinate model specification directly with the PI

Intelehealth | Hybrid (Baltimore & India)

Sep 2025 – Jun 2026

AI & Data Research Intern

- Designed a **stratified evaluation framework** (Cochran’s formula) to **benchmark LLM-generated outputs** against a **4,169-case gold standard**, quantifying accuracy and surfacing systematic **failure modes**

FOUNDER & VENTURE EXPERIENCE

Open Health Intelligence (ClarityAtlas) | Baltimore, MD

Sep 2025 – Present

Co-Founder & Technical Lead

- Designed and deployed a **tappigraphy-based passive sleep-estimation algorithm** inferring sleep windows from phone-interaction and motion signals; built the full stack end to end (Swift, SwiftUI, Supabase) and shipped it to a **live pilot at Johns Hopkins**
- Architected **HealthKit signal integration** (Core Motion, step count, gait speed) and a “Sleep Twins” matching service using **weighted Euclidean distance** over standardized behavioral features
- Received a non-binding ~\$75K **investment indication** (~7–8% **equity**); deferred the round to prioritize strategic domain partnerships
- Presented to **JHU President Ron Daniels** and the **JHU Data Science & AI Institute (DSAI) leadership**; invited to the **DSAI Undergraduate Committee**
- Selected for the **JHU Pava Spark Accelerator** and **NSF I-Corps**; presented at SF Tech Week and Penn HealthX

APPLIED ML PROJECTS

SnapSort | Phoenix, AZ

Jan 2024 – Aug 2025

Founder

- Trained a **convolutional neural network in PyTorch** for facial-landmark prediction, using **GridSearchCV** for hyperparameter search and model selection; reached ~95% **PCK**
- Built and shipped the surrounding AI photo-sorting product (Flask, AWS, **multi-model inference pipeline**) to **300+ users**

Gloria — AI Memory Companion | Hackathon Project

2025

Sole Developer

- Built an end-to-end **LLM application** with **OpenAI embeddings**, **Pinecone vector search**, a **Neo4j knowledge graph**, and a **prompt-driven agent state machine**

PUBLICATIONS & SELECT AWARDS

- Sole author, “Understanding the Effect of Agricultural Practices on Valley Fever through a Novel Statistical Model Using Dust Emissions,” *Journal of Student Research* (2024) — implemented an **SIR compartmental model** in Python from its differential equations
- Co-author, “Genomes of cressdnviruses, microviruses, a gyrovirus, and a caudovirus...,” *Microbiology Resource Announcements* (American Society for Microbiology, 2024)
- 63rd National JSHS Oral Presentation Finalist (top ~150 of 8,000+ nationwide)